NIKHIL KUMAR TIWARI

Full-Stack Developer | Computer Science (AI/ML) Student

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LinkedIn | Portfolio | GitHub

Professional Summary

Computer Science (AI/ML) student and Full-Stack Developer with experience building web applications using React, Next.js, Node.js, and Express.js. Hands-on experience with Python, Retrieval-Augmented Generation (RAG), vector embeddings, and LLM-powered applications. Interested in developing scalable web applications and integrating AI capabilities into practical software solutions.

Projects

Glucose Twin – AI-Powered Predictive Digital Twin GitHub | Demo

Next.js, React, Python, FastAPI, XGBoost, Pandas, CGMacros

- Built a healthcare dashboard to visualize glucose readings, model predictions, patient information, and risk indicators through interactive charts and a 3D digital twin interface.
- Worked on time-series data processing and predictive analytics for glucose forecasting and high/low glucose risk estimation using the project's trained models.
- Implemented replay-based simulation using historical CGMacros observations to demonstrate live monitoring while preserving original measurements and timestamps.

LeetCode Clone GitHub | Demo

Next.js, TypeScript, PostgreSQL, Prisma, Docker, Judge0, Better Auth

- Developed a full-stack coding platform with separate participant and admin interfaces, enabling users to solve programming problems and administrators to manage problems, test cases, examples, hints, and editorials.
- Integrated Judge0 for multi-language code execution, custom input testing, and evaluation against hidden test cases with execution results returned to the user.
- Implemented authentication and protected access using Better Auth, with PostgreSQL and Prisma for persistent data management and Docker-based Judge0 deployment for code execution.

Visitor Pass Management System GitHub | Live Demo

MongoDB, Express.js, React.js, Node.js, JWT, QR Code

- Built a full-stack visitor management system with dedicated admin and role-specific dashboards for managing visitor requests, approvals, pass issuance, and entry records.
- Implemented JWT authentication, role-based access control (RBAC), and protected routes to secure access to administrative and visitor management operations.
- Integrated QR code generation and camera-based scanning for visitor pass verification and entry tracking, along with automated email notifications and PDF visitor pass generation.
- Developed RESTful APIs and MongoDB data models to support visitor workflows, user management, and communication between the React frontend and Express backend.

Technical Skills

Languages: C, C++, JavaScript, TypeScript, Python
Frontend: React.js, Next.js, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, RESTful APIs, Microservices
Databases: PostgreSQL, MongoDB, Prisma ORM, Redis
AI / Machine Learning: Scikit-learn, PyTorch, RAG, LLMs, Vector Embeddings, LangChain
Tools Git, GitHub, Docker
Core Concepts: OOP, DBMS, Operating Systems, Computer Networks, System Design Fundamentals

Education

VIT Bhopal University Bhopal, India
Bachelor of Technology in Computer Science and Engineering (AI/ML) Expected Aug. 2029
CGPA: 8.37/10

Army Public School Jodhpur, India
Intermediate (Class XII) Aug. 2024
Final Grade: 83.4%

Certifications

- Tutedude – MERN Stack Development Course
- Complete React and Next.js Course with AI-Powered Projects